

Tanish Singh Rajpal

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EDUCATION

Carnegie Mellon University

MS in Information Technology – Information Security

Pittsburgh, PA

Graduating May 2027

Mukesh Patel School of Technology Management & Engineering

B.Tech in Computer Engineering – GPA: 3.83/4.00

Mumbai, India

Aug 2021 – May 2025

WORK EXPERIENCE

Newton's Apple

Vulnerability Assessment and Penetration Testing Intern

Pune, India

Oct 2023 – Jan 2024

- Executed penetration testing on a confidential oil company's Android application using Burp Suite and Metasploit, identifying and helping remediate 3 critical vulnerabilities that protected sensitive operational data.
- Performed static Java code analysis and collaborated with developers, leading to a 40% reduction in common security flaws in subsequent code releases.

RESEARCH & PUBLICATIONS

Rajpal, T. S., & Naithani, A. (2025). NeuroCrypt: A Neuro-Symbolic AI Ecosystem for Advanced Cryptographic Data Security and Transmission. *IEEE Transactions on Artificial Intelligence*. [Early Access]. <https://ieeexplore.ieee.org/document/11034683>

- Architected a neuro-symbolic AI model that uses a rule-based cryptographic engine with a deep neural network, dynamically generating symmetric keys to enhance encryption adaptability and resistance to cryptanalysis.
- Engineered and evaluated the system against classical ciphers (AES, DES) and neural-based attacks, demonstrating a 25% increase in security resilience and a 15% reduction in key management overhead.

Naithani, A., & Rajpal, T. S. (2025). Blockchain-Based Digital Health Passport: A Decentralized Framework for Secure and Interoperable Health Data Management. *IEEE Transactions on Information Technology in Biomedicine*.

<https://ieeexplore.ieee.org/document/11100281>

- Architected a decentralized health data management system using blockchain to ensure data immutability and patient sovereignty through a unique identifier and OTP-based consent mechanism, enabling secure, nationwide interoperability.
- Engineered a hybrid on/off-chain storage model integrated with public-key encryption and role-based access control, demonstrating enhanced security against unauthorized access and scalability for large medical records while maintaining regulatory compliance.

INDEPENDENT STUDY

Fidelity-Constrained Routing Algorithms for Heterogeneous Quantum Networks: A Multi-Path Analysis of Negative-Logarithmic and Parallelized Q-Learning Environments

Information Networking Institute, Carnegie Mellon University

May – Oct 2026

Advisor: Prof. Michael Mattarock

- Built a decentralized, PPO-based reinforcement learning router in NetSquid with a fixed, scale-invariant observation/action space, benchmarked against centralized negative-log-fidelity Dijkstra and hop-count baselines across both homogeneous and heterogeneous quantum hardware models.
- Achieved $\geq 99\%$ mean entanglement fidelity on homogeneous grids from 3×3 to 10×10 (90–93% under heterogeneous hardware noise), with a 100% routing success rate in five of six tested configurations — the sixth, 10×10 homogeneous, reached $84.4\% \pm 27.0\%$ across seeds — each result validated across 3 independent training seeds with a stochastic-evaluation robustness check.
- Research preprint and journal submission targeted for October 2026; simulation code, trained models, and results to be released as an open-source repository upon completion.

MASTER'S THESIS

A Hybrid Architecture for Post-Quantum EAP-TLS: Mitigating PMTUD Blackholes via Edge QUIC Proxies and QKD Backbones

Information Networking Institute, Carnegie Mellon University

Aug 2026 – Apr 2027

Advisors: Prof. Pedro Bustamante & Prof. Michael Mattarock | Reader: Prof. Patrick Tague

- Architecting and simulating (NS-3) a hybrid WPA-Enterprise 802.1X network that resolves silent Path MTU Discovery (PMTUD) blackholes caused by oversized post-quantum (ML-DSA) EAP-TLS certificates, by combining an edge QUIC-terminating proxy at the Wireless LAN Controller with a Quantum Key Distribution (QKD)-secured wired backbone.
- Designed four testable hypotheses spanning handshake latency (targeting a $\geq 40\%$ Flow Completion Time reduction under elevated wireless error rates), RADIUS server compute offload, IPv4/IPv6 baseline isolation, and sustained QKD key-delivery capacity under high-density authentication bursts.
- Thesis defense and final report targeted for Spring 2027; simulation source code and datasets to be released via a CMU-hosted GitHub repository and the CMU KiltHub data repository.

PROJECTS

Quantum Portfolio Optimization Using QAOA Team Lead	April 2026
Yale University's YQuantum 2026 Hackathon View repository	
• Led team to a 3rd place win by developing a quantum computing solution to optimize insurance investment portfolios using the Quantum Approximate Optimization Algorithm (QAOA). • Constructed the quantum circuit by mapping classical Markowitz financial models into a QUBO formulation and converting them into an Ising Hamiltonian.	

LEADERSHIP

Information Networking Institute (INI)	Pittsburgh, PA
<i>Student Ambassador</i>	<i>October 2025 – Present</i>
• Served as a student ambassador for the Information Networking Institute (INI), acting as a primary point of contact for prospective students to explain programs and campus life. • Supported the planning and execution of community events to foster engagement and strengthen the Carnegie Mellon University network.	

COMPETITIONS & CERTIFICATIONS

- **Completed IBM Qiskit Global Summer School (QGSS) 2026 – finished the full lab series (Labs 0–4) and the Advanced Track “Vault Challenge,” earning the Excellence Badge.**
- **3rd Place in Yale's premier quantum computing hackathon 2026.**
- 1st Place, Crypto Quest Cybersecurity Competition | Tech-Fest Ambiora 2024
- Have an in-depth knowledge of CCNA, CCNP SCOR, and CEH certifications.